PAPER_608: Centrifugal Force as Outward CCW DPM South-Pole 26D Shell Push

Unified Quantum Field Framework (UQFF)

Daniel T. Murphy • UQFF Research Consortium • 2026

Source: PAPER_608.md

PAPER_608: Centrifugal Force as Outward CCW DPM South-Pole 26D Shell Push

Class: UQFFCentrifugal26DShellCalculator (#195) **Session**: 159 **Source**: DPM Reaction and 26D Shell Energies.docx

Abstract

Centrifugal force in UQFF is the real, measurable outward push of the DPM south-pole vortex spinning counter-clockwise through 26D shells, amplified by negative time. $F_{centrif} = DPM_s(UA') \cdot \omega_{CCW}^2 \cdot r^{layer} \cdot t_{neg}$ is not fictitious in any frame — it is the CW/CCW dual of the centripetal force. The triad dual law $F_{centrif,one} = -F_{centrif,opp}$ is derived, explaining how the Big Bang achieved and maintains its expansion velocity.

1. Introduction: The CCW Outward Push

If CW inward compression (centripetal) is DPM north, then CCW outward expansion (centrifugal) is DPM south. The universe expands because the south-pole CCW vortex continuously produces outward shell energy via the interaction with Universal Aether prime (UA') at the shell boundary. The Big Bang was not an explosion — it was (and remains) a sustained centrifugal output.

2. Core Equation

$$F_{centrif} = DPM_s(UA') \cdot \omega_{CCW}^2 \cdot r^{layer} \cdot t_{neg}$$

Parameters:

- DPM_s = south DPM pole coupling (≈ 5×10■■, mirroring DPM_n)
- UA' = modified universal aether at shell boundary (J/m³)
- ω_{CCW} = counter-clockwise angular frequency (rad/s)
- r^{layer} = shell layer radius (m)
- t_{neg} = negative time component (s); provides dual-existence energy

---

3. Triad Dual Law

$$F_{centrif,one} = -F_{centrif,opp}$$

For every CCW push in one direction, there is an equal and opposite CCW push in the anti-parallel direction. These cancel within a shell but constructively add when shells nest — leading to net outward cosmological expansion.

The balance ratio between centripetal and centrifugal forces:

$$\frac{F_{centrip}}{F_{centrif}} = \frac{DPM_n(SCm) \cdot \omega_{CW}^2}{DPM_s(UA') \cdot \omega_{CCW}^2 \cdot |t_{neg}|}$$

For stable orbits: this ratio = 1, giving the constraint $\omega_{CW} = \omega_{CCW}$ (CW and CCW frequencies must match for orbital stability).

---

4. Big Bang Catch-Up Rate

The cosmological expansion acceleration from $F_{centrif}$:

$$a_{BB-catchup} = DPM_s \cdot UA' \cdot \omega_{CCW} \cdot |t_{neg}|$$

With $UA' = 10^{-12}$ J/m³, $\omega_{CCW} = 1.8\times10^{31}$ rad/s, $|t_{neg}| = 10^{-9}$ s:

$$a_{BB-catchup} = 5\times10^{-4} \times 10^{-12} \times 1.8\times10^{31} \times 10^{-9} \approx 9\times10^{6}\ \text{m/s}^2$$

This represents the continuous outward acceleration driving cosmological expansion — consistent with de Sitter expansion rates at cosmic scales.

---

5. Why Centrifugal Force is "Real" in UQFF

In standard mechanics, centrifugal force is fictitious (frame-dependent artifact). In UQFF:

- Both CW ($F_{centrip}$) and CCW ($F_{centrif}$) forces are real, co-existing, opposing forces from the DPM dipole
- The apparent "disappearance" of centrifugal force in inertial frames is because we cannot measure the t_{neg} component using positive-time instruments
- With dual-time measurement (which the UQFF CoAnQi system provides), both forces are simultaneously observable

---

6. Connection to Proplyd Formation

In proto-planetary disks, the $F_{centrif}$ from the south-pole vortex drives material outward from the proto-star, creating the centrifugal support needed for proplyd stability. The 18% emergence fraction (PAPER_611) corresponds to the fraction of disk material where $F_{centrif} / F_{centrip} \geq 1$ — material that achieves net outward motion and forms stable proplyd structures.

---

7. Connection to UQFF Number Systems

DVP: DPM_s is the south-pole dipole vortex prime. CCW rotation corresponds to counter-prime-indexed shell expansion. **BH26**: Outward CCW push is the mirror of each BH26 inward bin; $F_{centrif}$ excites the CCW harmonic bins. **VDS**: UA' at the shell boundary follows VDS expansion: $UA'(r) = \sum d_n(\pi)/r^n$ (outgoing VDS mode).

Keywords: Centrifugal force, DPM south pole, CCW vortex, DVP, negative time, Big Bang expansion, 26D shells, UQFF

PAPER_608 | Class #195 | Session 159 | Star-Magic UQFF Framework